Predicting Student Grade Trajectories

Evidence from Cortez and Silva (2008)

D200 – Machine Learning in Economics – Jacob Schwandt

Can background features predict student grade growth?

Motivation, Research Question & Hypotheses

The gap:

- Pritchard and Wilson (2003): social and economic background predicts student academic performance
- Cortez and Silva (2008): demographic and lifestyle features predict end-of-year grade levels (G3)
- Both predict where students end up → not their trajectory

Conditional on initial performance, can demographic and lifestyle features predict grade trajectory ($\Delta = G3 - G1$)?

H1 – Feature Validation

Background features carry genuine signal on starting performance.

H2 – Trajectory Null

Demographic and lifestyle features do not have predictive powers over grade trajectories.

H3 – Binary Direction

Conditional on the failure of continuous prediction in H2, background features may still discriminate between students who improve or hold steady versus those who decline.

Data & Outcome

What data do we use?

Dataset – Cortez and Silva (2008)

Sample construction

	Raw sample	G3 = 0 dropped	Analysed	Role
Maths	395	38	357	Primary dataset
Portuguese	649	15	634	Robustness check

G3 = 0 reflects exam non-participation, not zero attainment – including these would contaminate the trajectory outcome.

Features (30 total)

- Demographics: age, sex, address
- Family background: parental education & occupation
- Lifestyle: alcohol use, study time, going out frequency
- 13 numeric + 17 categorical (one-hot encoded)

Why Maths as primary?

- Higher trajectory variance: $\sigma = 1.50$ (Maths) vs 1.39 (Portuguese) – more signal to model
- Results replicated on Portuguese dataset as robustness check

What do we measure, and what leakage must we prevent?

Outcome Variable & Leakage

Outcome variables

- Continuous (H1 & H2): $\Delta = G3 - G1$ (grade trajectory)
- Binary (H3): $1[G_3 - G_1 \geq 0] = 1$ (improve/hold), 0 (decline)
- Before testing trajectory, validate features on G1 (levels) as proof-of-concept – a positive R^2 is required for H2 to be meaningful

Leakage risks

- G2 exclusion: recorded contemporaneously with G3 – including it predicts the outcome from itself
- Preprocessing leakage: StandardScaler and OneHotEncoder fitted on training data only, never on the full dataset

Preprocessing pseudo-code

```
# fit on train only
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test  = scaler.transform(X_test)

encoder = OneHotEncoder(
    handle_unknown='ignore',
    sparse_output=False
)
X_train = encoder.fit_transform(X_train)
X_test  = encoder.transform(X_test)

# WRONG leaks test distribution
scaler.fit(X_full)  # never do this
```

Regularisation

How are all regression models set up?

Regression Setup – Shared Across LASSO, Ridge, Elastic Net

Outcome & loss

- Target: $\Delta = G3 - G1$ (grade trajectory)
- Loss function: MSE – Gaussian MLE of conditional mean; appropriate for continuous outcome
- Null model: predict mean Δ for every observation

Evaluation metric

- R^2 relative to null model
- $R^2 < 0$ means model performs worse than predicting the sample mean – a meaningful benchmark
- Also report MSE for comparability

Hyperparameter tuning

- 5-fold GridSearchCV on training data only
- $\lambda \in [10^{-3}, 10^{1}]$ (50 log-spaced values)
- Best λ selected by lowest CV MSE

Train / test split

- 80 / 20 random split
- All preprocessing fitted on training fold only
- Test set used once: final evaluation only

Does LASSO find sparse trajectory signal?

LASSO – Primary Regression Model

Method

- Penalty: L1 ($\lambda\|\beta\|_1$) – applies soft-threshold operator, driving small coefficients to exactly zero
- Feature selection: built-in – natural for 30 features where signal is likely sparse
- Sparsity interpretation: retained features are genuinely informative; zeroed features are irrelevant

Results

- H1 (G1 levels): $R^2 = 0.150$ – feature set carries signal on starting performance
- H2 (Δ trajectory): $R^2 = 0.057$ – 3 features retained (age, absences, freetime)
- Retained features have small, diffuse coefficients – no single feature dominates

LASSO pipeline

```
from sklearn.linear_model import Lasso
from sklearn.pipeline import Pipeline
from sklearn.model_selection import GridSearchCV

pipe = Pipeline([
    ('scaler',  StandardScaler()),
    ('encoder', OneHotEncoder(
                  handle_unknown='ignore',
                  sparse_output=False)),
    ('lasso',   Lasso(max_iter=10000)),
])

param_grid = {
    'lasso__alpha': np.logspace(-3, 1, 50)
}
grid = GridSearchCV(pipe, param_grid,
                    cv=5,
scoring='neg_mean_squared_error')
grid.fit(X_train, y_train)
```

Do robustness checks alter the conclusion?

Ridge & Elastic Net – Robustness Checks

Ridge

- Penalty: L2 ($\lambda\|\beta\|^2_2$) – uniform shrinkage, never zeroes coefficients
- Feature selection: none – dense shrinkage benchmark
- Result: R^2 = -0.024 – worse than predicting the mean
- Implication: when signal is sparse, dense regularisation fails to reduce variance sufficiently. Validates LASSO sparsity assumption.

Elastic Net

- Penalty: L1 + L2 – interpolates between LASSO and Ridge via mixing parameter α
- $\alpha \in \{0.1, 0.3, 0.5, 0.7, 0.9, 1.0\}$ tuned jointly with λ by CV
- Result: $R^2 \approx$ LASSO across all α – closely tracks sparse solution
- Implication: signal structure is sparse, not grouped. Adding L2 mixing does not recover signal that LASSO misses.

Three regularisation strategies converge on near-zero R^2 for trajectory. This rules out model misspecification as the explanation for the null result.

Is grade trajectory predictable?

Results – Feature Validation (H1) & Trajectory Regression (H2)

H1: Feature validation

LASSO on G1: $R^2 = 0.150$ (MSE = 8.03) – features carry genuine signal on starting performance. The null result in H2 is therefore meaningful, not a feature-set artefact.

H2: results

Model	R^2	Retained features
LASSO	0.057	age, absences, freetime
Ridge	-0.024	All (below null)
Elastic Net	≈ LASSO	Tracks LASSO across l1_ratio
Portuguese (robustness)	≈ 0	Confirms null result

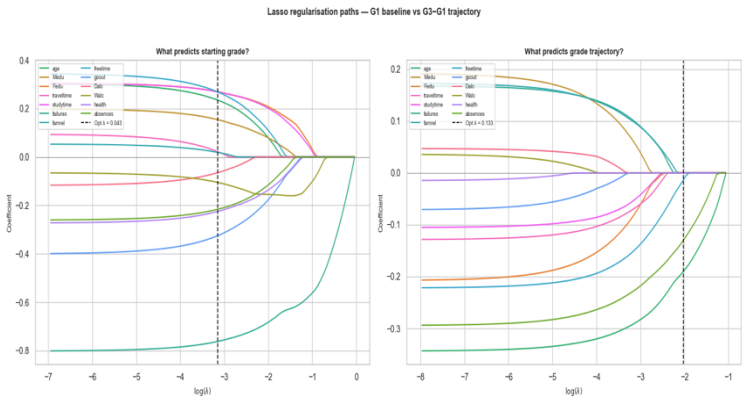

Figure 2: LASSO regularisation paths — G1 baseline vs G3−G1 trajectory

- x-axis: log(λ) – penalty increases right to left
- Most coefficient paths remain at zero across full range
- Only 3 features survive at optimal λ: age, absences, freetime

Classification

Can we predict direction if not magnitude?

Classification Methods – L1 Logistic & Random Forest

Transition: Regression R^2 near zero – the continuous trajectory is not predictable. Reframe: can we predict the direction of change? Outcome becomes $1[\Delta \geq 0]$: improve/hold (1) vs decline (0).

L1 Logistic Regression

- Penalty: L1, solver: SAGA – consistent with LASSO sparsity assumption
- Loss: cross-entropy – calibrated probability estimates
- class_weight='balanced' – reweights by inverse class frequency (66:34 imbalance)
- $C = 1/\lambda$, CV-tuned on training data
- Metric: ROC-AUC – invariant to class balance

Random Forest

- Non-linear benchmark: tests whether trajectory signal exists beyond linear models
- n_estimators $\in \{100, 200, 300\}$; max_depth $\in \{3, 5, 10, \text{None}\}$; min_samples_leaf $\in \{1, 5, 10\}$
- class_weight='balanced'
- Decorrelated ensembling: each split samples $m < p$ features — trees make different errors, bagging reduces variance
- Post-hoc: SHAP TreeExplainer for feature attribution

Does binary classification outperform a random classifier?

Results – H3: Binary Direction

0.609

L1 Logistic AUC

absences only retained

0.682

Random Forest AUC

diffuse attribution

0.500

Random classifier

baseline

SHAP Mean Absolute Attributions (Random Forest)

Feature	SHAP value
Absences	0.027 ←
Age	0.010
Freetime	0.010
Father's education	0.009
Going out	0.009

Endogeneity of absences:

Declining students may disengage → accumulate absences. Model may capture reverse causality – policy intervention on absences may not break the causal chain.

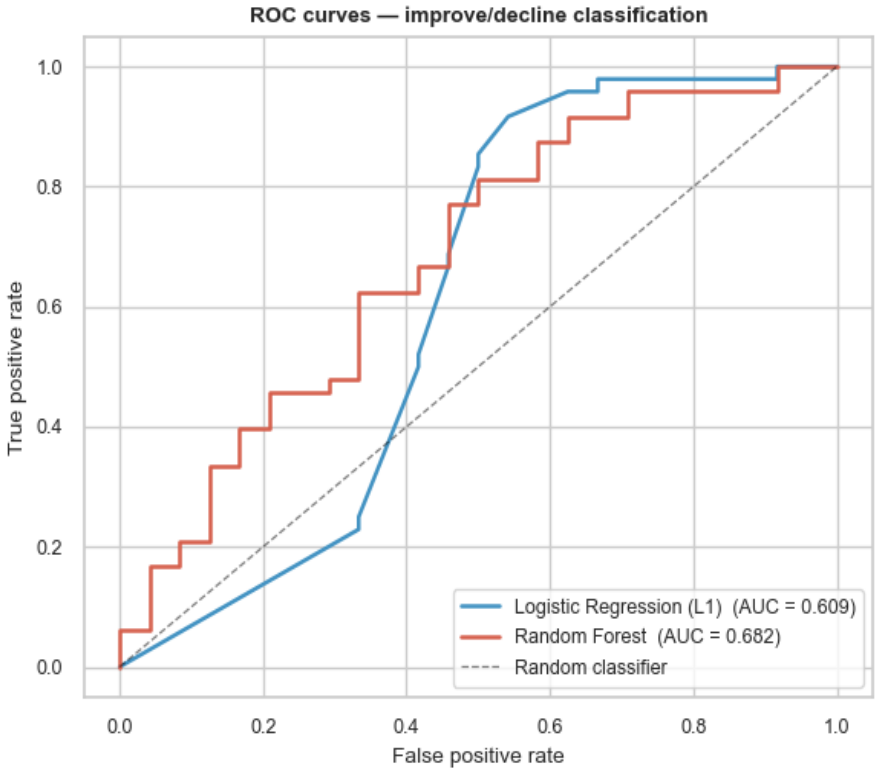

Figure 3: ROC curves – L1 Logistic (0.609) vs Random Forest (0.682)

Discussion & Extension

What do the results tell us about predictability?

Discussion

The H1/H2 gap – levels vs trajectory

- Background features predict where students begin, not how much they grow
- Conditional on G1, socioeconomic sorting variables lose predictive power entirely
- Trajectory driven by within-student dynamics that observable characteristics do not capture

Endogeneity of absences

- Absences dominate SHAP but are plausibly endogenous
- Declining students disengage → absences accumulate
- Model may capture reverse causality
- Absences are a symptom, not a lever – policy intervention may not break the causal chain

Non-linearity (Random Forest)

- AUC gain of 0.682 vs 0.609 suggests mild non-linear interactions
- Signal exists but not structured around policy-actionable features
- Non-linearity exists; it is simply not the right kind

What are the limits, and where could this go?

Challenges & Extensions

Challenge	Extension
Small n – 357 maths students, two schools only	Larger longitudinal administrative dataset
Feature set misses teacher quality & peer effects	Richer data: teacher fixed effects, classroom resources
G1 conditioning – potential collider bias	Instrumental variables or panel fixed effects
Absences likely endogenous	Lagged absences, Granger style causality tests
Binary threshold (≥0) is arbitrary	Ordered outcome model; continuous trajectory

Central limitation:

The null result in H2 cannot distinguish between true unpredictability of trajectory and feature-set limitations. A richer dataset may overturn the finding.